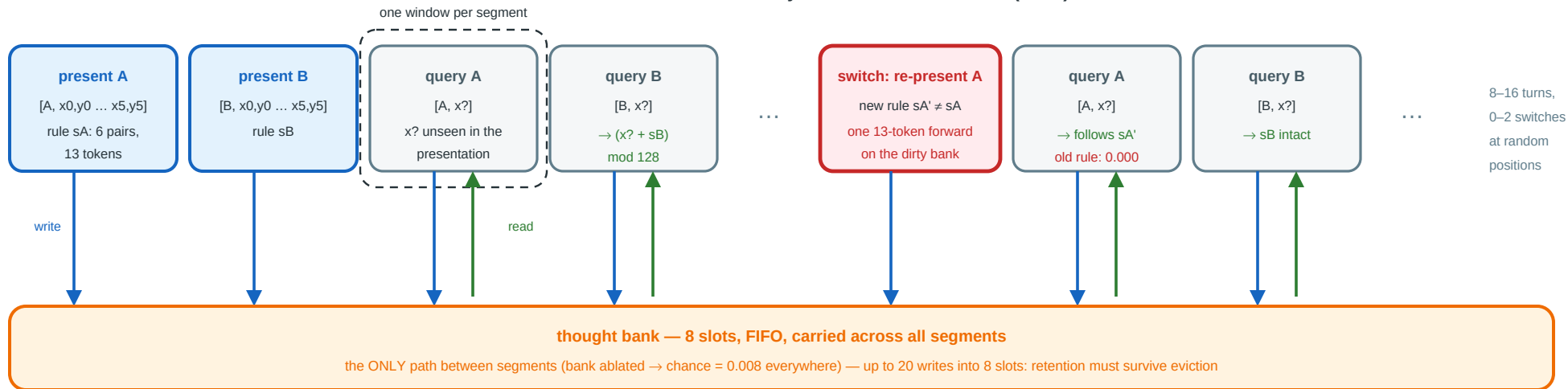


one conversation of the keyed fresh-rule benchmark ($K = 2$)



rule family

$y = (x + s) \bmod 128$, one fresh offset s
per key per conversation

train / held split on s

multiples of 8 (15 rules) never trained;
a held rule at test = a never-trained binding
inside the meta-learned family

what makes it memory

information shown once, used many turns
later, never resolvable from the current
window; queries use unseen $x \rightarrow$ the
binding must generalize, not be recalled

structure randomization (the policy trainer): length $\sim U[8,16]$, up to two switches at random positions — a random key is re-presented with a new rule on the carried bank.

S8 shows this distribution, and nothing architectural, is what installs the memory policy (keep / overwrite / write-on-dirty).